



CE 222 – Computer Organization and Assembly Language (COAL)

Lecture 26

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Lecture Outline

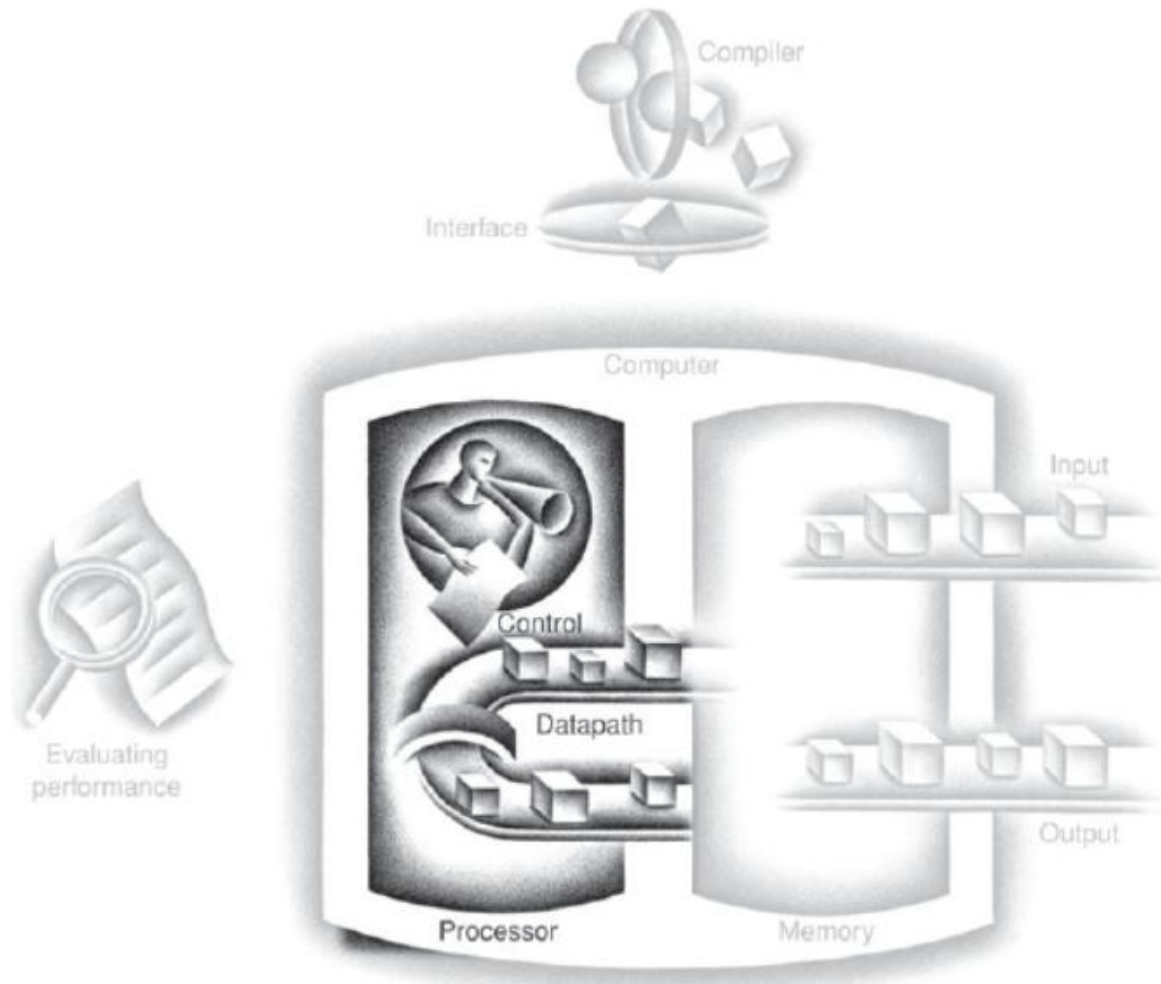
- Announcements (Assignment and Quiz 3)
- Outline of Previous Lecture:
 - **Chapter 4 – The Processor**
 - 4.2 Logic Design Conventions
 - 4.3 Building a Datapath
- Outline of Today's Lecture:
 - **Chapter 4 – The Processor**
 - 4.3 Building a Datapath
 - 4.4 A Simple Implementation Scheme

Chapter 4

The Processor

Chapter Theme

The Five Classic Components of a Computer



Chapter Outline

Chapter 4 – The Processor

- 4.1 Introduction
- 4.2 Logic Design Conventions
- 4.3 Building a Datapath
- 4.4 A Simple Implementation Scheme
- 4.5 Multicycle Implementation
- 4.6 An Overview of Pipelining
- 4.7 Pipelined Datapath and Control
- 4.8 Data Hazards: Forwarding versus Stalling
- 4.9 Control Hazards
- 4.10 onwards (self-reading)

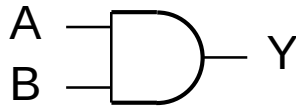
Logic Design Basics

- Information encoded in binary
 - Low voltage = 0, High voltage = 1
 - Wire vs. **Bus**
- **Combinational** element
 - Example processor unit
- **State (sequential)** elements
 - Example processor unit

Combinational Elements

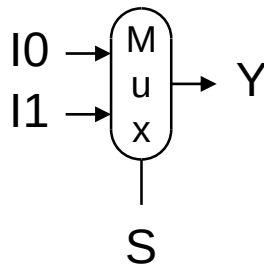
- AND-gate

- $Y = A \& B$



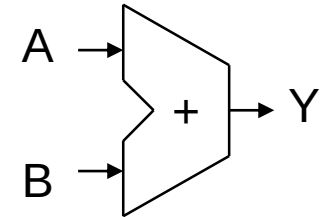
- Multiplexer

- $Y = S ? I1 : I0$



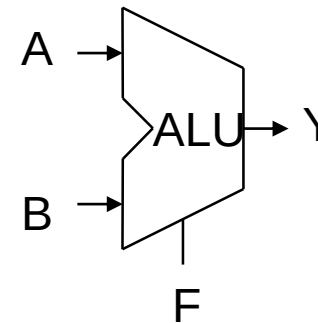
- Adder

- $Y = A + B$



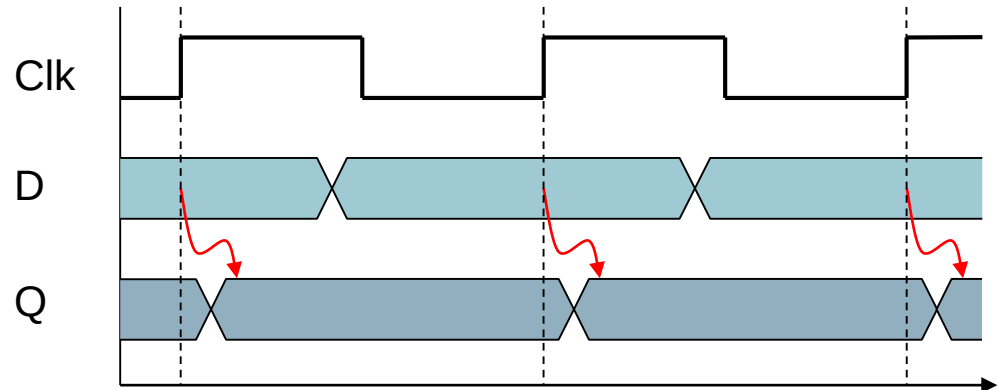
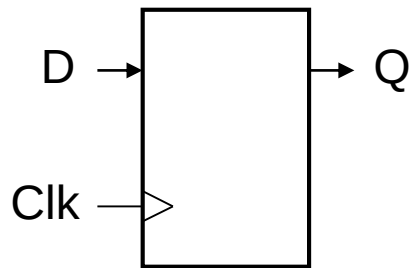
- Arithmetic/Logic Unit

- $Y = F(A, B)$



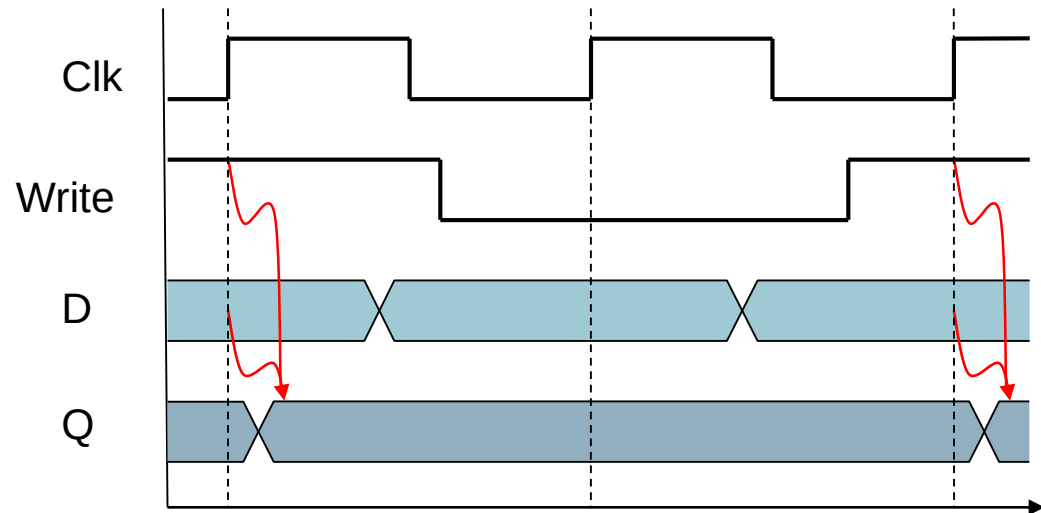
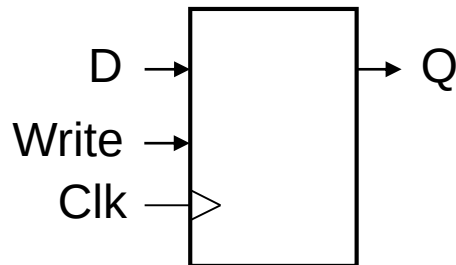
Sequential Elements

- Register: stores data in a circuit
 - Edge-triggered



Sequential Elements

- Register with **write control**
 - Only updates on clock edge when write control input is 1
 - Used when stored value is required later

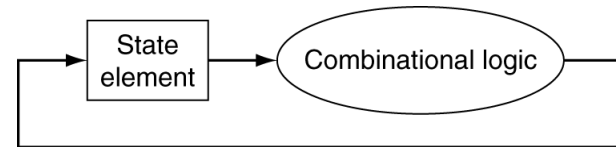
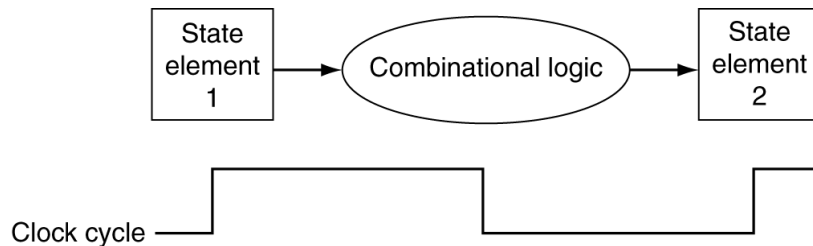


Clocking Methodology

- A clocking methodology (e.g. edge-triggered) is required/ designed to make hardware predictable
- Asserted vs. de-asserted signal

Clocking Methodology

- Combinational logic generally present between state elements
 - Combinational logic transforms data during clock cycle i.e. between clock edges
 - Longest delay determines clock period
 - Implementation impacts clock cycle!



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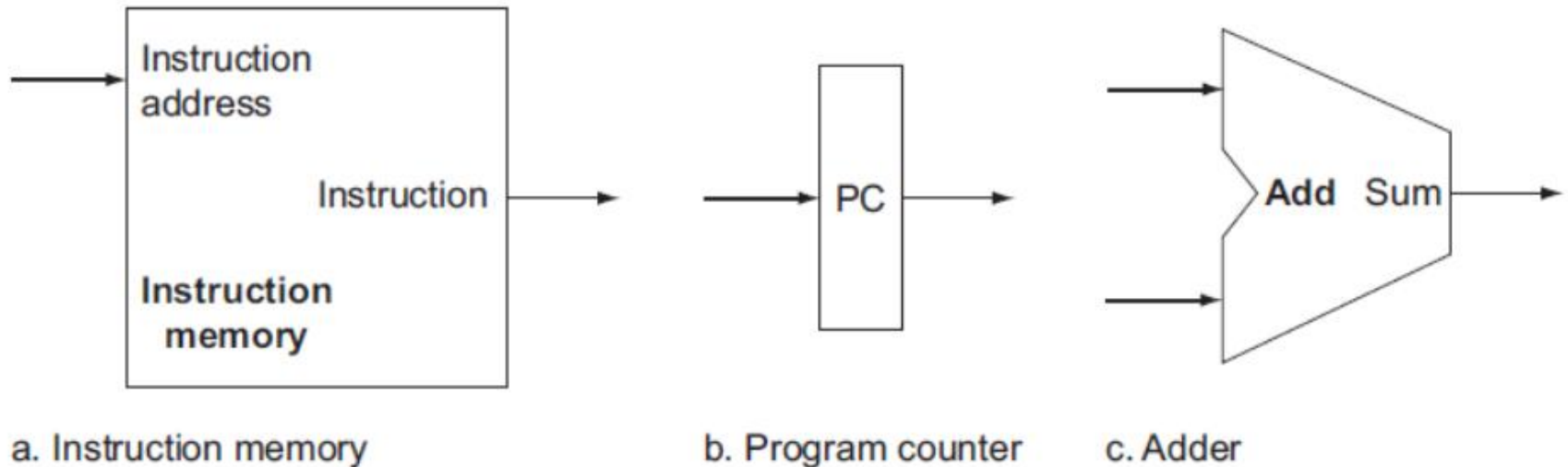
Building a Datapath

- Datapath
 - Set of elements that process data and addresses in the CPU
 - Registers, ALUs, mux's, memories, ...
- We will build a RISC-V datapath incrementally
 - Refining the overview design

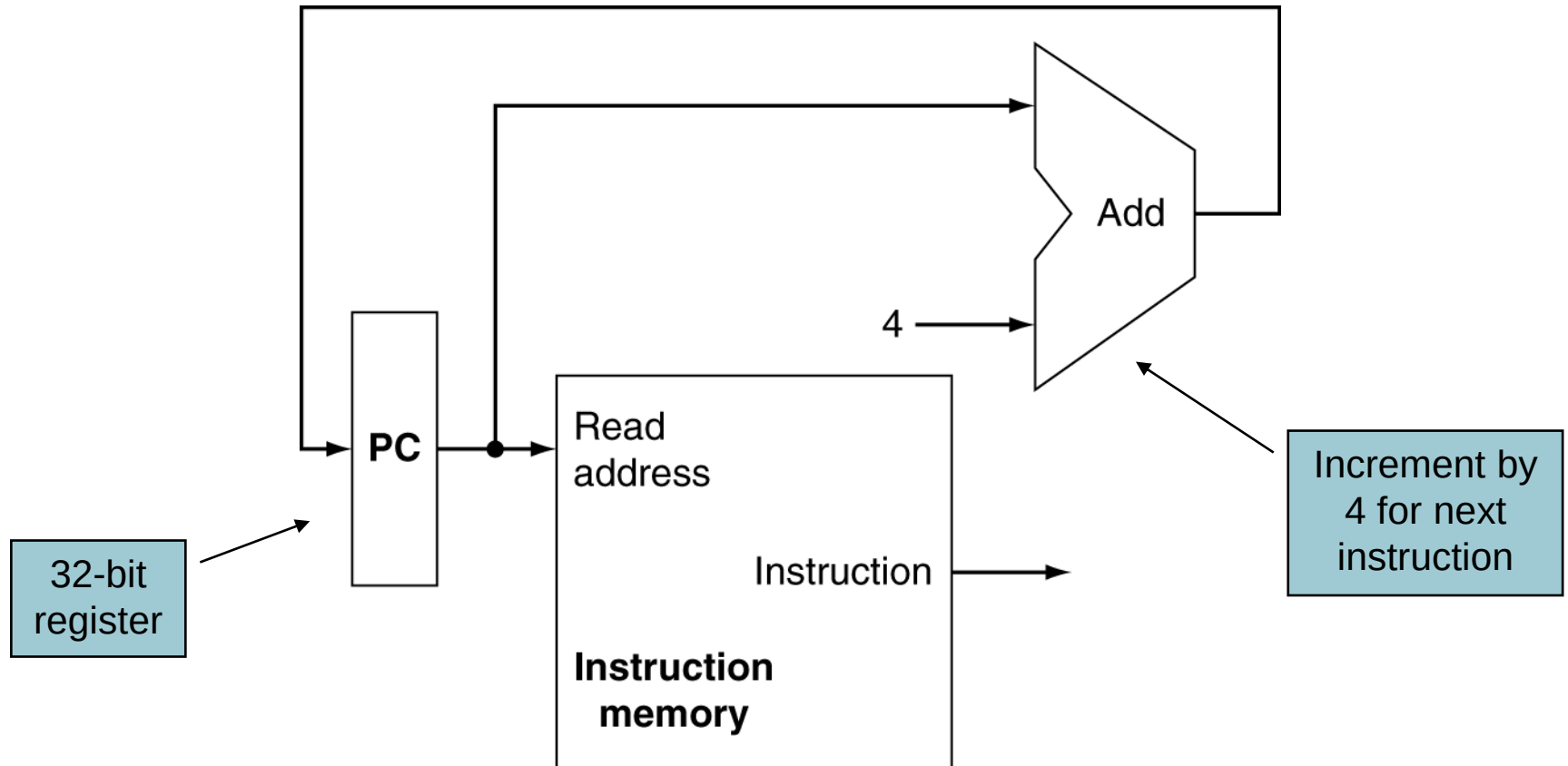
Datapath for Instruction Fetch

Datapath Elements

- PC width

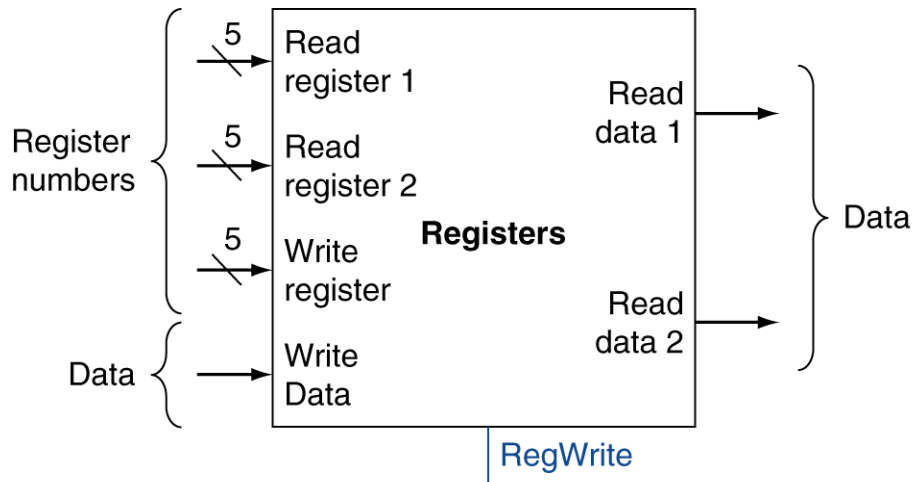


Datapath for Instruction Fetch

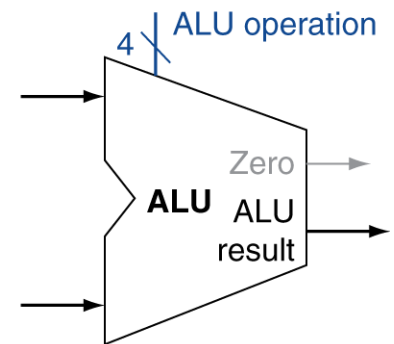


R-type Instructions

- Steps in executing R-type instructions:
 1. Read two register operands
 2. Perform arithmetic/logical operation
 3. Write result in register-file
- I/Os for the register-file



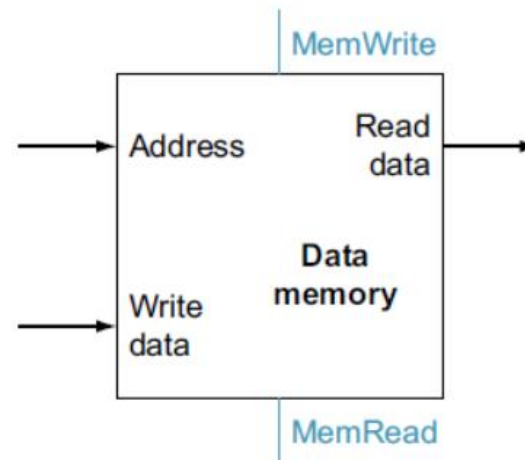
a. Registers



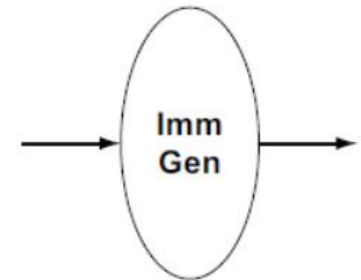
b. ALU

Memory access Instructions

- Steps in executing memory-access (load, store) instructions:
 1. Read register operands
 2. Calculate address using 12-bit offset
 - Use ALU, but sign-extend offset
 3. Load: Read memory and update register
 4. Store: Write register value to memory
- I/Os for the datapath elements

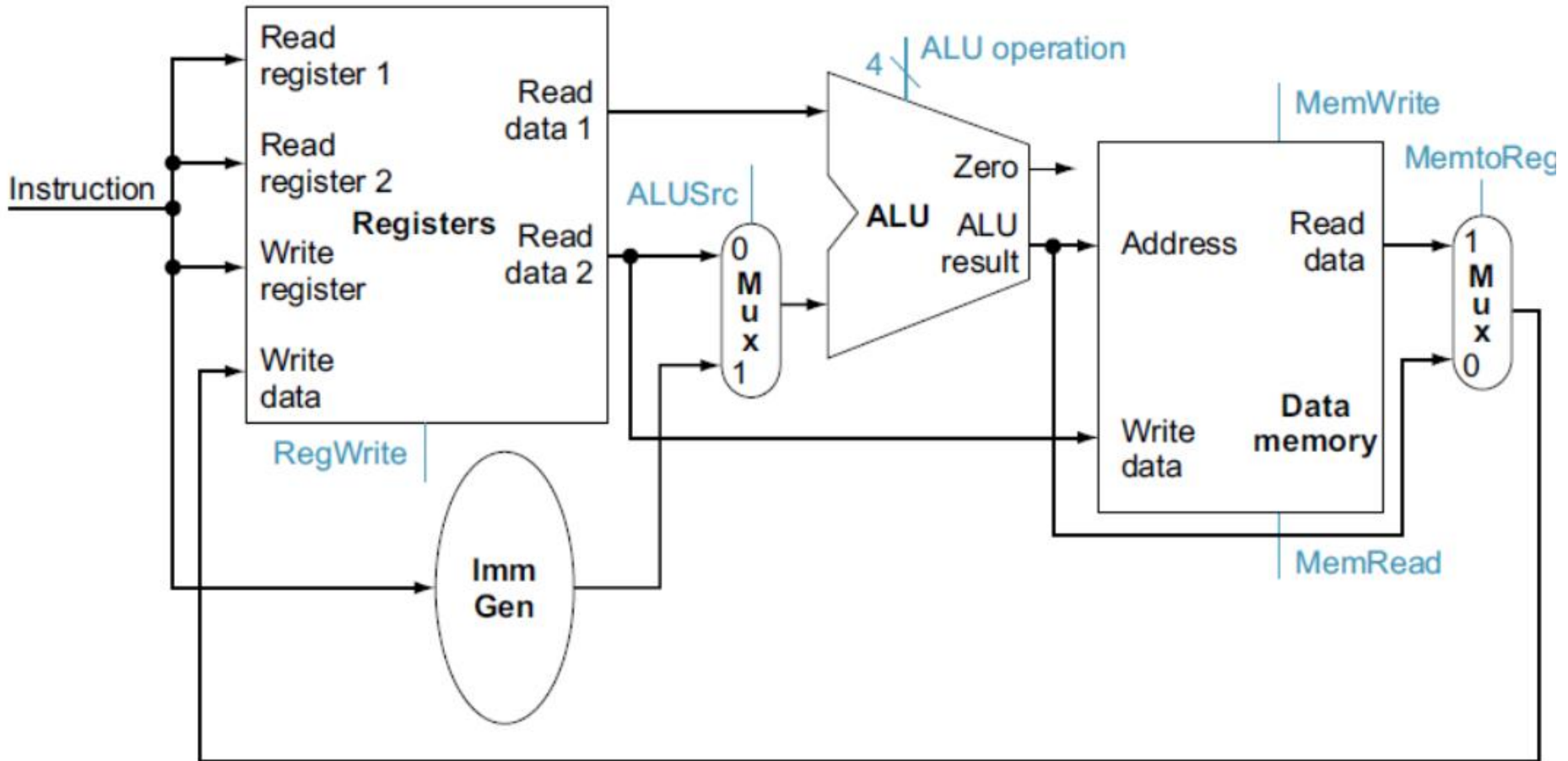


a. Data memory unit



b. Immediate generation unit

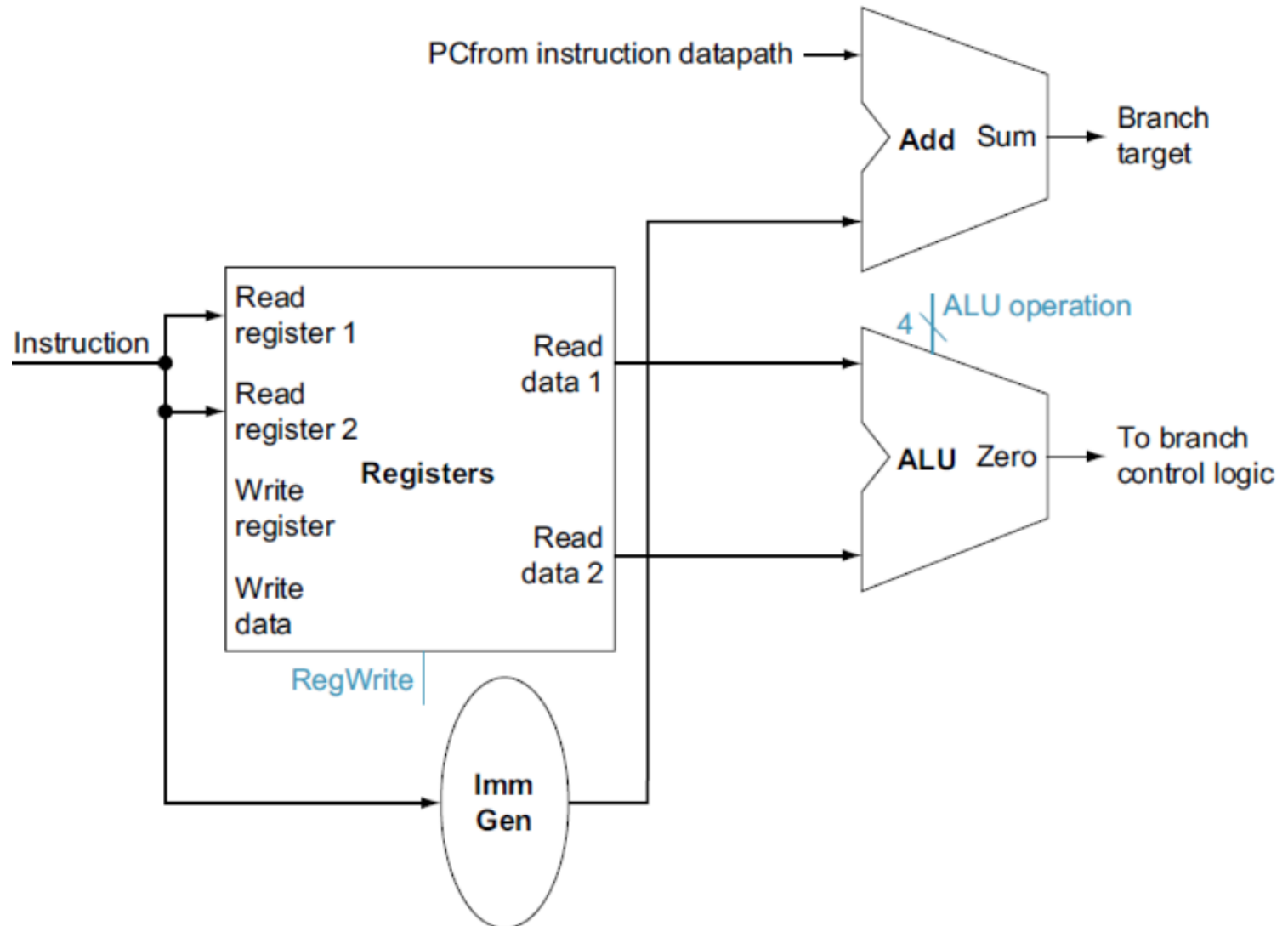
R-type + Memory-access Instructions datapath



Branch Instructions

- Steps in executing branch instructions:
 1. Read register operands
 2. Compare operands
 1. Use ALU, subtract and check Zero output
 3. Calculate target address
 - Sign-extend offset
 - Shift left 1 place (halfword displacement)
 - Add to PC value

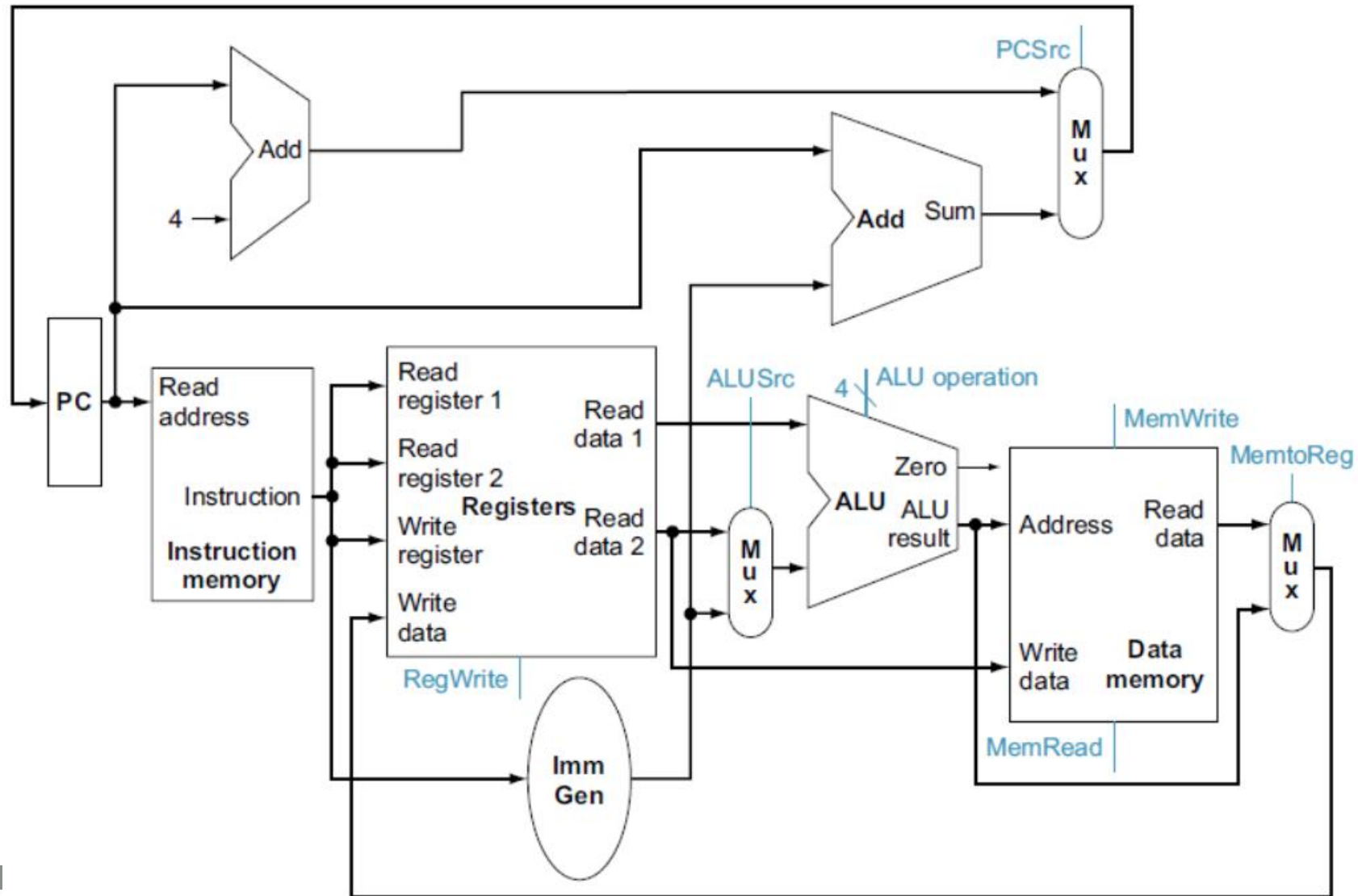
Branch Instructions datapath



Integrated datapath

- Integrating datapaths for instruction fetch, arithmetic/R-type, memory-access, branch instructions into a single datapath
- Use multiplexers where alternate data sources are used for different instructions

Integrated Datapath



Integrated datapath

- Single-cycle processor
 - All steps for an instruction must happen in the right sequence -> each datapath element can only do one function at a time
 - We need separate instruction and data memories
 - This data path executes an instruction in one clock cycle

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